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Development of high-performance microfluidic stabilizers via particle swarm optimization strategy

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ABSTRACT

Microfluidic stabilizers effectively suppress flow fluctuations and are, therefore, essential for peristaltic pump-driven microfluidic systems and high-precision instrumentation. Air-chamber-based passive microfluidic stabilizers are structurally simple, but their performance depends on multiple geometric parameters, making customized design challenging. To address these issues, this study proposes a particle-swarm-optimization-based optimization strategy to engineer the key structural parameters of a microfluidic stabilizer according to specific demands. A theoretical dynamic model is established to elucidate the stabilization mechanism and identify the key parameters governing device performance. Global optimization of the geometric parameters is then performed using particle swarm optimization, yielding an improved design with enhanced stabilization capability and practical manufacturability. The optimized device was fabricated and evaluated through instantaneous flow measurements at different flow rates and in intracranial pressure drainage simulations and droplet generation experiments. The results show that over the tested wide flow-rate range, the device achieves a flow pulsation suppression rate of more than 95%, reduces the flow fluctuation rate to less than 0.2%, and effectively attenuates pulsations above 0.1 Hz. These results confirm that the proposed optimization strategy effectively improves both stabilization performance and manufacturability, providing a complete design framework that integrates theoretical modeling, particle swarm optimization-based global optimization, and experimental validation for the rapid development of customized microfluidic stabilizers.

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I. INTRODUCTION

Microfluidic systems have been widely applied in biology,[1,2] medicine,[3,4] and chemistry[5,6] due to their advantages of high integration, high sensitivity, high precision, and low cost. In many of these applications, flow conditions directly affect system performance.[7,8] Peristaltic pumps are widely used in microfluidic systems because they enable continuous, non-contact fluid delivery with simple operation.[9,10] However, the operation of peristaltic pumps relies on rollers, which can induce periodic fluctuations in flow rate and pressure.[11] To mitigate pump-induced flow pulsations and achieve stable flow output, both active and passive flow stabilization strategies have been developed. Active methods usually rely on external sensing, power modules, or control algorithms,[12–14] which increase system complexity and cost. By contrast, passive stabilizers do not require additional external control units and offer advantages such as simple structure, ease of integration, and portability, making them more suitable for miniaturized microfluidic applications.

Passive stabilizers can generally be classified into membrane-based and air-chamber-based designs. Although membrane-based stabilizers can effectively buffer flow pulsations, their long-term performance is often constrained by material fatigue, elasticity degradation, and limited pressure tolerance.[15–18] Consequently, air-chamber-based stabilizers have attracted increasing attention as simpler and potentially more robust alternatives.[19,20] Recent studies have explored various air-chamber-based stabilization strategies.

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Zhou *et al.* utilized standing air bubbles as hydraulic capacitors, established a theoretical model based on Euler's equation, and achieved a 75.3% reduction in syringe pump-induced flow pulsations.[21] Zhang *et al.* developed an on-chip peristaltic pump integrating two air chambers as flow stabilizers, reducing the instantaneous flow variation to within 3%.[22] More recently, a Lego-like modular microfluidic stabilizer system with tunable stabilization ratios was demonstrated using 3D printing, enabling multi-level stabilization combinations through the control of fluidic RC constants.[23] Despite these advances, however, the quantitative relationships between key geometric parameters and stabilization performance remain insufficiently characterized, and systematic structural optimization of these parameters has not been performed.

Systematic optimization of the structural parameters of the microfluidic stabilizer often requires numerical optimization techniques. Among the commonly used methods in microfluidics, simulated annealing (SA) has been applied to chip layout optimization,[24] and genetic algorithms (GA) have been employed for micromixer geometry design.[25] These methods each have their own applicable scope and tuning requirements: the performance of SA is closely related to the cooling schedule, while GA involves coordinated tuning of multiple hyperparameters. In contrast, particle swarm optimization (PSO) is a derivative-free, population-based global optimization method.[26] It offers fast convergence, ease of implementation, and requires few control parameters, making it particularly attractive for design optimization when the objective function can be evaluated efficiently.

To address the aforementioned issues of existing air-chamber-based microfluidic stabilizers, this study proposes a strategy to develop customized fluid stabilizers using PSO. A theoretical model of the air-chamber-based stabilization mechanism is first established on the basis of Euler's equation to clarify the underlying mechanism. On this basis, global optimization of the structural parameters is carried out according to the customized demands. After determining the optimized parameter combination, the device is fabricated and its performance is verified, achieving a flow pulsation suppression rate of more than 95%, reducing the flow fluctuation rate to less than 0.2%, and effectively attenuating pulsations above 0.1 Hz. In addition, application demonstrations, such as intracranial drainage and droplet generation, further demonstrate that the stabilizer can effectively attenuate pressure disturbances and enable more stable and precise liquid delivery.

The main contributions of this work are threefold. First, a theoretical dynamic model is established based on Euler's equation to elucidate the stabilization mechanism of air-chamber-based microfluidic stabilizers, revealing that the device behaves as a first-order low-pass filter. Second, a variance-based Sobol global sensitivity analysis is used to objectively quantify the parameter hierarchy, revealing that the gas-chamber radius and the main-channel height carry the largest global effects, while the side-channel radius serves as the key frequency-tuning parameter; building on this structure, PSO efficiently determines the optimal parameter values, which simplifies the design workflow. Third, the optimized device is experimentally validated across a wide range of flow rates and in two practical application scenarios: intracranial pressure drainage and droplet generation. The results demonstrate significant suppression of flow and pressure pulsations. The design and optimization strategy proposed in this work demonstrates its effectiveness in enhancing the performance and manufacturability of fluid stabilizers. Moreover, it provides a systematic route for exploring the design space and identifying the key parameters that govern device performance, thus offering a useful tool for the community when developing customized microfluidic systems and prototypes.

II. THEORETICAL ANALYSIS AND OPTIMIZATION PROCESS

A. Principle of the microfluidic stabilizer

To facilitate the designing of air-chamber-based stabilizers, we propose a strategy to customize device parameters using particle swarm optimization (PSO). The proposed microfluidic stabilizer is a passive device that exploits the compressibility of gas in a sealed chamber to attenuate flow pulsations. As shown in Fig. 1(a), the stabilizing effect is achieved through a gas–liquid coupling mechanism. A single chamber buffers transient inlet-flow fluctuations, while a multistage series configuration further strengthens pulsation attenuation and stabilizes the outlet flow. To describe the transient response of the stabilizer, a theoretical dynamic model was established for the air-chamber-based microfluidic system, and the detailed derivation is presented in the supplementary material. The ratio of the fluctuation amplitude at the outlet of the microfluidic stabilizer to that at the inlet is defined as η and can be expressed as

$$\eta = \left(\frac{\varepsilon}{\varepsilon + 1}\right)^{n}, \tag{1}$$

$$\varepsilon = \frac{\left(\frac{\rho l_{1,n}}{A_{1,n}} s + R_{1,n}\right)}{\left(\frac{\rho l_{2,n}}{A_{2,n}} s + R_{2,n} + \frac{p_{2,n}}{V_{2,n} s} + \frac{\rho g}{A_{2,n} s}\right)}. \tag{2}$$

As indicated by Eq. (1), the dynamic behavior of the proposed microfluidic stabilizer can be approximated as that of a first-order low-pass filter. In this system, the flow resistance determines the damping characteristics, while the gas-volume compliance provides the buffering effect, allowing effective attenuation of flow fluctuations above the cutoff frequency.

B. PSO algorithm and process

Based on the mathematical model established above, the main-channel length l_1, the main-channel width w, the main-channel height h, side-channel length l_2, side-channel radius r, and gas-chamber radius R were selected as the optimization variables, denoted by X, because they govern the equivalent flow resistance and dynamic response of the system.

Particle swarm optimization (PSO) is a population-based stochastic optimization technique inspired by the social foraging behavior of bird flocks and fish schools. In PSO, each candidate solution is represented as a particle that moves through the search space. The movement of each particle is guided by its own historical best position and the global best position found by the entire swarm, allowing the algorithm to efficiently explore the solution space and converge to optimal solutions with relatively few control parameters.[27]

The PSO algorithm is adopted to globally optimize X under multiple constraints. A comparative study of PSO with genetic algorithm (GA) and simulated annealing (SA) is provided in

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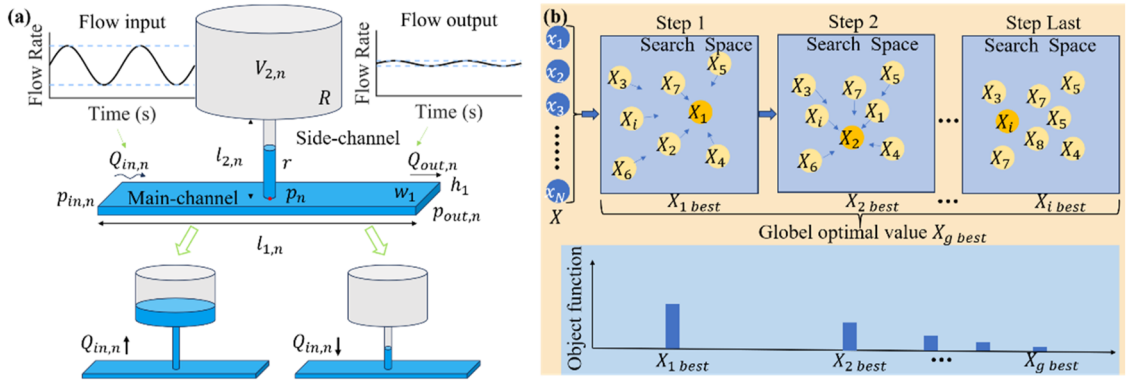


FIG. 1. Process of optimizing structural parameters. (a) Device principles and structural parameters. (b) Schematic view of the PSO process. X represents the input optimization parameters; $X_{i\ best}$ represents the local optimum; and $X_{g\ best}$ represents the global optimum. The objective function tends to reach a target value after a series of iterations.

the supplementary material. The dependence of convergence on the number of optimization variables is also analyzed in the supplementary material. Based on fabrication capabilities and device design considerations, the search ranges were determined, as listed in Table I. The lower bounds of h, r, and w are set by the minimum feature sizes achievable with micromilling. The upper bounds of h, l_1, l_2, and R are limited by flow resistance and the overall chip footprint, with R also constrained by l_1 to maintain a compact layout. Since a smaller value of η indicates better flow stabilization performance, the objective function is defined to minimize η over the frequency range of 0–30 Hz, which covers the typical pump-induced pulsation frequencies in microfluidic systems.[28–30] In this optimization framework, each candidate solution $X = (l_1, w, h, l_2, r, R)$ generated by PSO is evaluated by calculating its corresponding η using the theoretical model expressed in Eqs. (1) and (2). The value of η serves as the fitness function that guides the search process: particles with lower η are considered better and influence the subsequent updates of particle positions and velocities. The PSO algorithm then iteratively updates the candidate solutions based on this fitness information until the optimal parameter combination is found.

Figure 2(a) shows the convergence behavior of the PSO algorithm for structural parameter optimization, with representative pulsation frequencies of 1, 5, 10, 15, 20, and 25 Hz. The optimal stabilization ratio is generally approached within approximately ten population-equivalent batches (about 400 objective-function evaluations, including initialization; Table S5), demonstrating that the method can efficiently determine the optimal parameter combination and the corresponding stabilization performance at each frequency. The rapid early improvement is consistent with the smooth objective function, the pronounced trends along several influential parameter directions, and the relatively broad low-value regions observed in the conditional slices. Accordingly, only a small number of swarm updates are needed to enter the low-value region, after which additional iterations yield no further improvement. A detailed analysis of the objective-function smoothness and convergence behavior is provided in the supplementary material.

Figure 2(b) shows the optimized structural parameters at different pulsation frequencies, revealing the influence of chamber geometry on the dynamic stabilization performance of the microfluidic stabilizer. For pulsation suppression at a specific frequency, a corresponding optimal parameter combination can be obtained. Over the frequency range of 0–30 Hz, the main-channel length l_1, side-channel length l_2, the main-channel width w, gas-chamber radius R, and the main-channel height h vary only slightly, whereas the side-channel radius r shows more pronounced variations. To investigate the influence of these parameters on η, Sobol sensitivity analysis was performed and the details are provided in the supplementary material. The results show that R and h have the largest global effects on the amplitude ratio, while r becomes increasingly influential at higher frequencies but contributes little at low frequencies. The optimization results shown in Fig. 2(b) further show that the optimal values of R and h remain stable over 1–30 Hz, whereas r exhibits the most pronounced variation. Therefore, l_1, l_2, w, R, and h are finally set to 20, 5, 1, 4 mm, and 100 μm, respectively, while the side-channel radius, r, which exhibits the most pronounced variation with frequency, is selected as the primary parameter for frequency-dependent tuning.

Figures 2(c) and 2(d) characterize the theoretical stabilization performance of the optimized device over the frequency range of 0–30 Hz in terms of the stabilization ratio and suppression ratio, respectively. As the frequency increases, the theoretical optimal stabilization ratio decreases markedly, while the theoretical

TABLE I. Search ranges of the optimization variables.

Parameters	Search range (mm)
h	(0.1, 0.2)
l_1	(5, 20)
w	(1, 3)
l_2	(4, 5)
r	(0.2, 0.5)
R	(2, 4)

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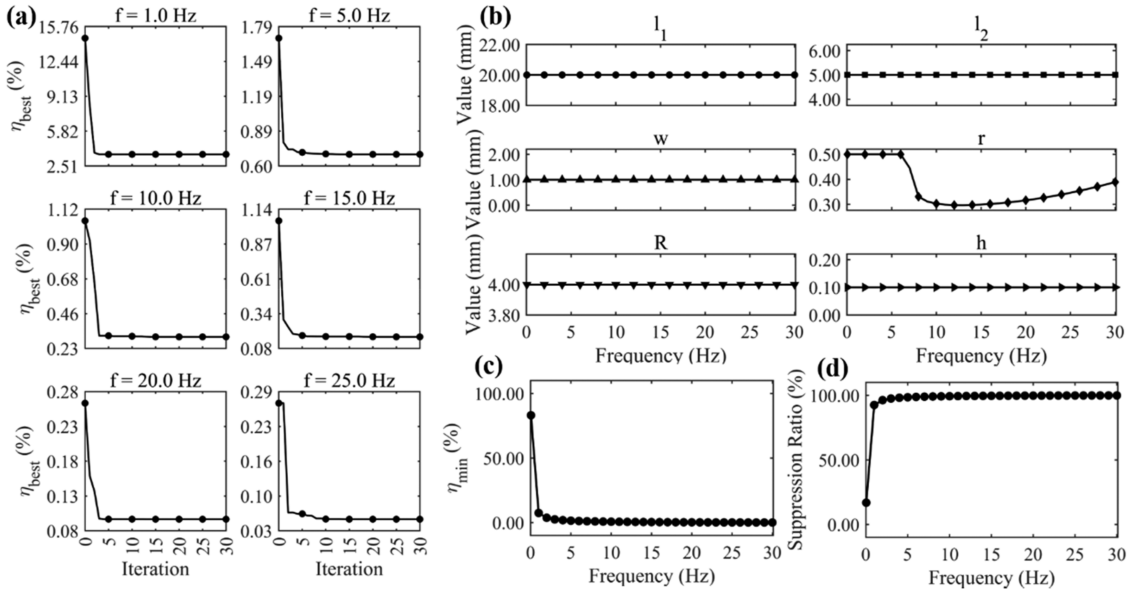


FIG. 2. Optimization of the proposed microfluidic stabilizer over the target frequency range using the PSO algorithm. (a) Evolution of the optimization process with the iteration number at frequencies of 1, 5, 10, 15, 20, and 25 Hz. (b) Optimized structural parameters over the frequency range of 0–30 Hz. (c) Theoretical optimal stabilization ratio η_{min} as a function of pulsation frequency. (d) Theoretical optimal suppression ratio as a function of pulsation frequency. The suppression ratio represents the percentage reduction in flow fluctuation amplitude after passing through the stabilizer.

optimal suppression ratio rises rapidly and approaches 100%, indicating that the device exhibits a clear low-pass filtering behavior and provides stronger attenuation when dealing with medium- and high-frequency pulsations within the target frequency range. These results further confirm the effectiveness of parameter optimization in improving the broadband dynamic stabilization performance of the microfluidic stabilizer.

C. Microfluidic stabilizer design and fabrication

The optimization results show that r exhibits the most pronounced variation with frequency, whereas l_1, l_2, w, R, and h remain nearly constant. As revealed by the Sobol sensitivity analysis in the supplementary material, R and h have the largest global effects on the amplitude ratio, but their optimal values remain stable across the frequency range; in contrast, r becomes increasingly influential at higher frequencies. The strong frequency dependence of r can be understood from Eq. (2), where the side-channel flow resistance scales as $1/r^4$, making r the most sensitive frequency-tuning parameter. By contrast, the other geometric parameters primarily govern the global amplitude, and their optimal values remain stable over the target frequency range. The mean flow rate does not appear in the transfer function because the model is linearized around the mean flow and thus it was not treated as an optimization parameter. The value of r can be adjusted according to the specific stabilization requirements of practical applications. Based on the above-mentioned analysis, the suppression-ratio curves corresponding to r = 0.5 mm and r = 0.4 mm are plotted, as shown in Fig. 3(a). Compared with r = 0.4 mm, the parameter r = 0.5 mm provides better suppression performance in the low-frequency region, with the suppression ratio at 1 Hz increasing by up to 21.7%. In the high-frequency region, however, the suppression ratio decreases by only about 1%. Therefore, considering both the improvement in low-frequency suppression and the retention of high-frequency performance, r = 0.5 mm is more suitable as the final design parameter for the device.

By substituting the optimized structural parameters into the mathematical model, Fig. 3(b) shows the transient flow responses for different numbers of series stages, n. Without the microfluidic stabilizer, the input flow exhibits pronounced periodic pulsations. Introducing a single stabilizing chamber significantly reduces output fluctuations. As n increases from 1 to 4, the output flow gradually approaches its mean value, and the fluctuation amplitude decreases continuously. A one-stage to four-stage microfluidic stabilizer reduces the input flow fluctuation amplitude from 100% to ~5.904%, 0.349%, 0.021%, and 0.001%, respectively.

To validate the designed microfluidic stabilizer, a laminated PMMA device consisting of four structurally identical stabilizing chambers connected in series by microchannels was designed, as shown in Fig. 3(d). The top and bottom layers serve as unpatterned sealing layers, while the two middle layers constitute the microchannel layer and the air-chamber layer, respectively. The microchannel layer contains the main and side channels, whereas the air-chamber layer contains sealed air chambers connected to the side channels. The device was fabricated by an external workshop using standard CNC machining on PMMA sheets. The main channel was micromilled using a 0.2 mm diameter carbide end mill at a spindle

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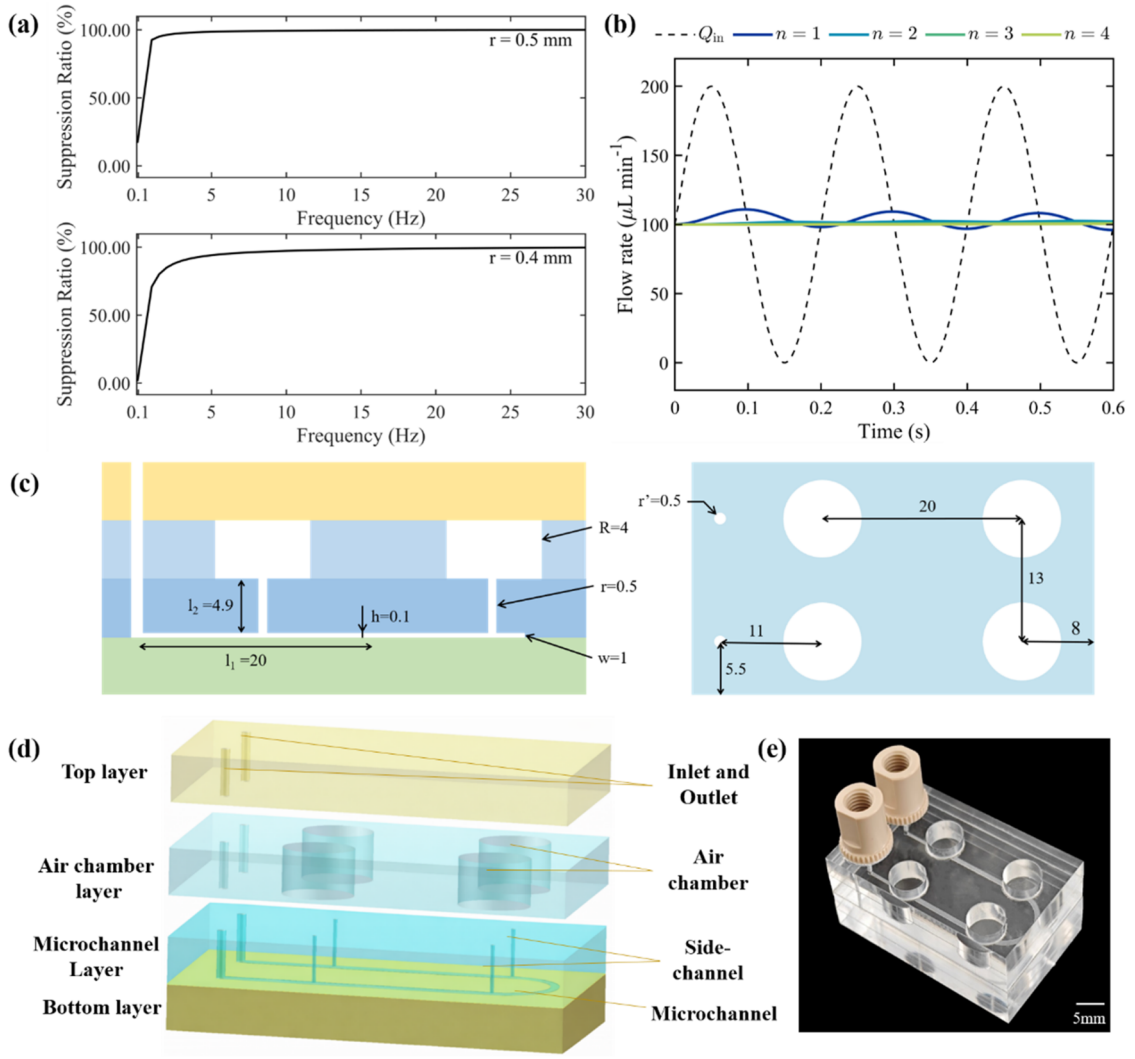


FIG. 3. Design and fabrication of a customized microfluidic stabilizer. (a) Theoretical suppression rate curves for the two sets of optimization parameters (r = 0.5 mm and r = 0.4 mm). (b) Flow-rate responses for different numbers of stabilizing chambers connected in series (n = 1–4), where Q_{in} denotes the inlet flow rate. (c) Dimensioned layout of the microfluidic chip with key geometric parameters (mm) indicated. (d) Layered structure of the microfluidic stabilizer. (e) Photograph of the fabricated stabilizer.

speed of 24 000 rpm and a feed rate of 10 mm/s, with a depth of cut of 20 μm per pass. The air chambers were drilled using an 8 mm diameter drill bit on a CNC engraving machine. The four PMMA layers were then thermally bonded at 100 °C under ~1 MPa for 10 min. The chip geometry is shown in Fig. 3(c). The final device measured 24 × 42 × 20 mm^3, with the inlet and outlet connected to external tubing through standard interfaces, as shown in Fig. 3(e).

III. EXPERIMENTAL TESTING AND APPLICATION VALIDATION

A. Performance evaluation

To evaluate the flow pulsation suppression performance of the proposed microfluidic stabilizer, a peristaltic pump (LabN1, Shenchen, Shanghai, China) was used as the flow source and the fabricated stabilizer was installed at the pump outlet. The flow signal was measured in real time using a flow sensor (MFS54, Elveflow, France). The flow rates were set to 100–800, 1000, and 2000 μl min^{-1} for testing. The severity of flow fluctuations was quantified by the instantaneous flow fluctuation rate, defined as the maximum absolute deviation of the instantaneous flow rate from the mean flow rate divided by the mean flow rate.

As shown in Fig. 4(a), the output of the peristaltic pump exhibited pronounced periodic pulsations without stabilization, particularly at low flow rates, whereas the flow became markedly smoother after the stabilizer was introduced. The FFT results shown in Fig. 4(b) confirm effective suppression of flow fluctuations over a wide frequency range, consistent with the low-pass filtering behavior

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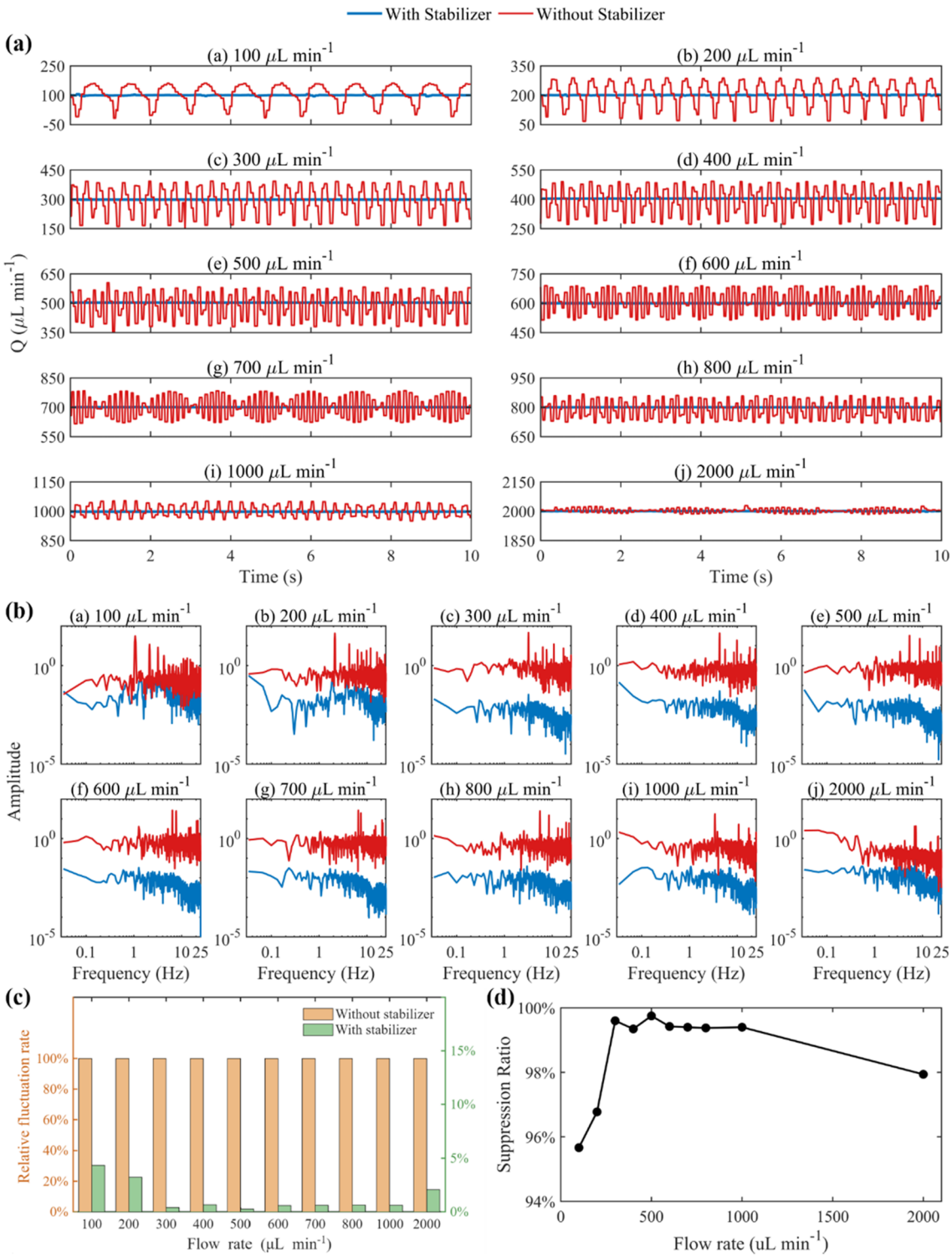


FIG. 4. Flow stabilization performance of the proposed microfluidic stabilizer at flow rates of 100–800, 1000, and 2000 μl min^{-1}. (a) Comparison of outlet flow rate as a function of time with and without the stabilizer. (b) Frequency-domain spectra of the outlet flow rate with and without the stabilizer. (c) Relative fluctuation rate at different flow rates, normalized to the unstabilized case (100%). (d) Suppression rate of flow pulsations as a function of flow rate.

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of the device. As quantified in Fig. 4(c), after normalizing the unstabilized fluctuation to 100% for each flow rate, the stabilizer reduces the fluctuation to less than 5% of the corresponding unstabilized value across all tested conditions. As shown in Fig. 4(d), the suppression rate remains above 95% over the entire flow-rate range of 100–2000 μl min^{-1} and exceeded 99% in the intermediate range of 300–1000 μl min^{-1}.

These experimental results are consistent with the theoretical analysis presented above. In particular, the raw output of the peristaltic pump exhibits more pronounced low-frequency periodic pulsations at low flow rates. After the introduction of the stabilizer, the fluctuation amplitude is significantly reduced over the entire tested range, consistent with the low-pass filtering behavior predicted by the model. In addition, the theoretical analysis indicates that a multistage series configuration can further enhance pulsation attenuation, and the experimental results confirm that the proposed structure effectively suppresses pump-induced periodic fluctuations. Taken together, these observations support the proposed PSO designing strategy and the performance of the microfluidic stabilizer device.

B. Application validation

To evaluate the pressure pulsation suppression performance of the optimized microfluidic stabilizer, an active fluid pumping system for intracranial drainage was constructed, as shown in Fig. 5(a).

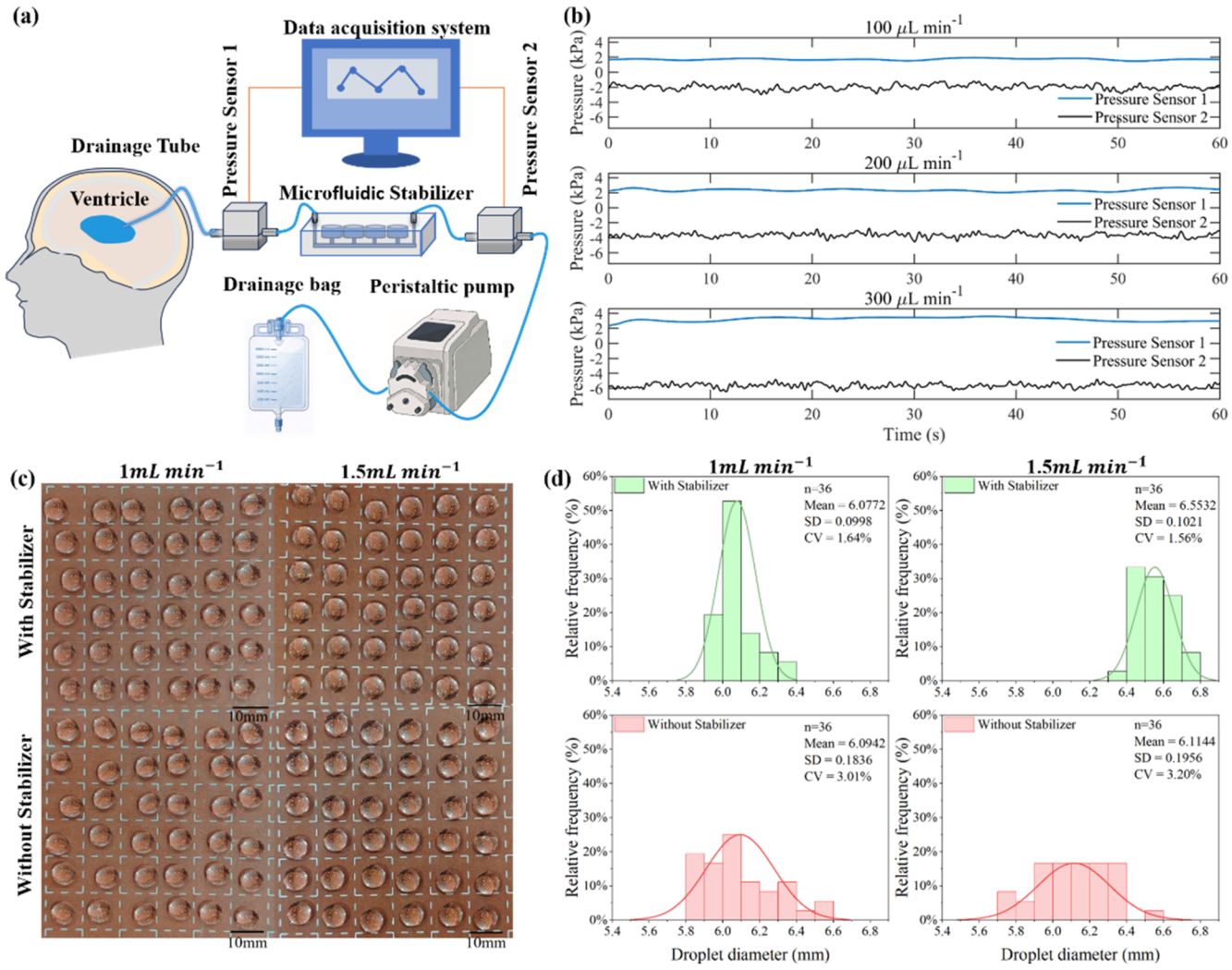


FIG. 5. Application validation of the proposed microfluidic stabilizer in different fluid control systems. (a) Schematic diagram of the intracranial pressure drainage setup incorporating the microfluidic stabilizer. (b) Pressure responses measured at flow rates of 100, 200, and 300 μl min^{-1}. (c) Comparison of droplet generation images with and without the microfluidic stabilizer at flow rates of 1.0 and 1.5 ml min^{-1}. (d) Statistical distribution of droplet diameters corresponding to panel (c). Data are presented as mean ± SD (n = 36 for each group), and the coefficient of variation (CV) was used to evaluate droplet uniformity.

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Normal human intracranial pressure is ~2 kPa, and cerebrospinal fluid drainage is required when the pressure rises abnormally. Existing active drainage systems commonly use a peristaltic pump as a non-contact driving element for intracranial pressure regulation.[31] Connecting a passive microfluidic stabilizer in series between the pump and the load provides a simpler and more direct way to attenuate pressure disturbances.

In the experiment, a simulated ventricular pressure was maintained at 2 kPa, and two pressure sensors (Lifemeta™ SPC, Cobetter, Hangzhou, China) were placed on either side of the microfluidic stabilizer for pressure measurement. Pressure sensor 1 was placed on the ventricle side and pressure sensor 2 was placed on the pump side, so that the pressure disturbance transmitted across the stabilizer could be directly assessed. The experimental results are shown in Fig. 5(b). Within the clinically relevant drainage flow-rate range of 100–300 μl min^{-1}, the pressure signal measured by pressure sensor 2 exhibits pronounced fluctuations, indicating significant pressure disturbances introduced by the operation of the peristaltic pump. By contrast, the pressure measured by using pressure sensor 1 remains substantially smoother and maintains effective suppression of pressure pulsations over the entire tested flow-rate range. These results demonstrate that the proposed microfluidic stabilizer can effectively buffer pressure disturbances generated by the peristaltic pump and reduce their transmission to the ventricle side, thereby improving the pressure stability of the intracranial pressure drainage system. In addition, the compact device structure facilitates serial integration into microfluidic systems.

The performance of the stabilizer was further evaluated in droplet generation experiments using a peristaltic pump as the liquid driving source. As shown in Figs. 5(c) and 5(d), the introduction of the microfluidic stabilizer narrowed the droplet size distribution at both 1.0 and 1.5 ml min^{-1}, reducing the coefficient of variation from 3.01% to 1.64% and from 3.20% to 1.56%, respectively. These results further confirm that the stabilizer suppresses flow pulsation transmission and improves output uniformity in practical microfluidic applications.

IV. CONCLUSION

In summary, we present a PSO-based structural design strategy for a passive microfluidic stabilizer, which provides a complete design framework integrating theoretical modeling, global optimization, fabrication, and experimental validation to suppress flow and pressure pulsations in pump-driven microfluidic systems. Theoretical analysis and optimization reveal that the stabilization performance is jointly governed by the hydraulic resistance of the microchannel geometry and the buffering capacity of the air chamber. This understanding enables identification of the parameters with the largest global effects and selection of an optimized four-stage configuration, in which the side-channel radius serves as the primary frequency-tuning parameter. In addition, experiments demonstrate that the optimized stabilizer effectively attenuates pump-induced disturbances over a wide flow-rate range, improves pressure stability in an intracranial drainage system, and improves droplet generation uniformity in a peristaltic-pump-driven microfluidic platform. Compared with conventional stabilization approaches that often rely on complex structures or external control, the proposed strategy provides a simple and readily integrable route for improving flow stability in microfluidic systems and may facilitate broader applications in hydrodynamic focusing,[32] cell sorting,[33] droplet generation,[34] and particle synthesis.[35]

SUPPLEMENTARY MATERIAL

The supplementary material contains the theoretical derivation of the dynamic model, a comparative study of the optimization algorithms, analysis of the dimensionality dependence of convergence, an objective-function smoothness and convergence assessment, and Sobol sensitivity analysis. First, a dynamic model of the air-chamber-based microfluidic stabilizer is established. Derived from Poiseuille's law and Euler's equation, the ratio of the fluctuation amplitude at the outlet of the microfluidic stabilizer to that at the inlet is obtained. Second, the comparison between the PSO algorithm with the genetic algorithm (GA) and simulated annealing (SA) in terms of computational efficiency and optimization stability is described. Finally, variance-based Sobol global sensitivity analysis was performed for the six geometric parameters.

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AUTHOR DECLARATIONS

Conflict of Interest

The authors have no conflicts to disclose.

Author Contributions

Hao Sun: Data curation (lead); Methodology (lead); Validation (lead). **Rencong Mu**: Data curation (equal); Methodology (equal); Validation (equal). **Mengxi Wu**: Conceptualization (equal); Methodology (equal); Writing – review & editing (equal). **Junshan Liu**: Project administration (lead); Validation (supporting); Writing – review & editing (supporting).

DATA AVAILABILITY

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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